Brazil then wants to buy a genetic algorithm for its health-care system, it would have a strong incentive to opt for the much more accurate Chinese algorithm rather than the one from New Zealand. If the Chinese algorithm then hones itself on more than 200 million Brazilians, it will get even better. Which would prompt more countries to choose the Chinese algorithm. Soon enough, most of the world's medical information would flow to China, making its genetic algorithm unbeatable.

The attempt to concentrate all information and power in one place, which was the Achilles heel of twentieth-century totalitarian regimes, might become a decisive advantage in the age of AI. At the same time, as noted in an earlier chapter, AI could also make it possible for totalitarian regimes to establish total surveillance systems that make resistance almost impossible.

Some people believe that blockchain could provide a technological check on such totalitarian tendencies, because blockchain is inherently friendly to democracy and hostile to totalitarianism. In a blockchain system, decisions require the approval of 51 per cent of users. That may sound democratic, but blockchain technology has a fatal flaw. The problem lies with the word 'users'. If one person has ten accounts, she counts as ten users. If a government controls 51 per cent of accounts, then the government constitutes 51 per cent of the users. There are already examples of blockchain networks where a government is 51 per cent of users.[7]

And when a government is 51 per cent of users in a blockchain, it has control not just over the chain's present but even over its past. Autocrats have always wanted the power to change the past. Roman emperors, for example, frequently engaged in the practice of *damnatio memoriae* – expunging the memory of rivals and enemies. After the emperor Caracalla murdered his brother and competitor for the throne, Geta, he tried to obliterate the latter's memory. Inscriptions bearing Geta's name were chiselled out, coins bearing his effigy were melted down, and the mere mentioning of Geta's name was punishable by death.[8] One surviving painting from the time, the Severan Tondo, was made during the reign of their father – Septimius Severus – and originally showed both brothers together with Septimius and their mother, Julia Domna. But someone later obliterated Geta's face and even smeared excrement over it. Forensic analysis identified tiny pieces of dry shit where Geta's face should have been.[9]